

Khulna University of Engineering & Technology
 Department of Urban and Regional Planning
 BURP 3rd Year 2nd Term Regular Examination, 2017
CE 3207
 (Elements of Civil Engineering Structures)



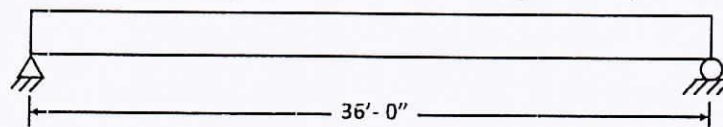
Full Marks: 210

Time: 3 hrs

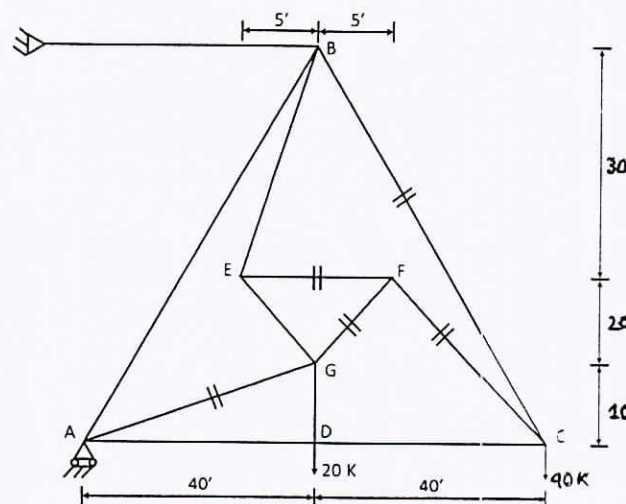
- N.B.
- i) Figures in the right margin indicate full marks.
 - ii) Assume reasonable value for any missing data.
 - iii) Answer any three questions from each section in separate script.

Section – A

- 1 For the simply supported beam shown in the following figure, determine the maximum stresses at midspan section due to its own weight and the following cases of loading and prestressing: (35)
- (a) A uniform live load of 500 lb/ft.
 - (b) A uniform live load of 500 lb/ft and an axial centroidal longitudinal compressive force of $P = 259.2$ K.
 - (c) A uniform live load of 2500 lb/ft and an eccentric longitudinal compressive force of $P = 259.2$ K acting at an eccentricity $e = 4$ in.
 - (d) A uniform live load of 2733 lb/ft and an eccentric longitudinal compressive force of $P = 259.2$ K acting at the maximum practical eccentricity for this section.
 - (e) The maximum live load when $P = 259.2$ K acting at $e = 6$ in.
- Use $b = 12$ in, $h = 24$ in, $f'_c = 4500$ psi and an allowable $f'_c = 2050$ psi.



- 2
- (a) Define Class U, Class T and Class C prestressed concrete flexural members. (06)
 - (b) Is it possible to design a prestressed concrete beam with no shear carried by the concrete section? Justify your answer. (07)
 - (c) Prove that, the stress in concrete due to the prestressing force after immediate losses can be equalled to the stress in transformed section due to the initial prestress. (07)
 - (d) Is suitable control on deflection very essential? Justify your answer. What are the factors influencing deflection? (07)
 - (e) Prove that the average loss in the prestressing cables is equal to the half of the loss of the first cable. (08)
- 3
- (a) Why should we analyze a truss? What are the different methods for analysis of a truss? When a truss member is called zero force member? (06)
 - (b) (29)

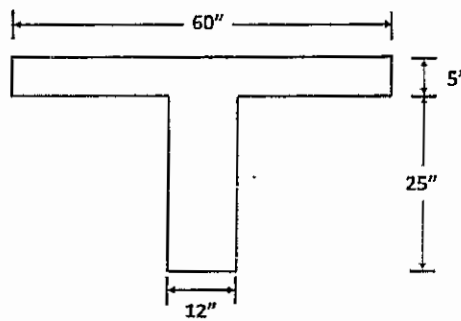


Define Braced Frame. Determine the forces in members AG, BC, EF, FC and FG of the truss as shown in figure below.

4. (a) Define the following terms: (08)
 (i) Design earthquake, (ii) Design pressure, (iii) Soft Storey, (iv) Storey drift
 (b) Calculate the design wind load at each floor level for a 5 storied building from the following data: (13)
 Plan area: 15m×10m, height of each floor = 5m, $C_1 = 1.5$, exposure category = B, $V_b = 200$ mph, $C_p = 1.5$, $C_z = 0.5212 z^{0.2841}$, $G_z = 1.46 z^{-0.067}$.
 (c) Calculate the earthquake load at each floor level for a 5 storied steel moment resisting frame building from the following data: height of each floor = 3m, plan area = 20m×30m, seismic dead load = 2000 kN/1st floor and 2500 kN/other floor, structural importance coefficient = 1.25, response modification coefficient = 10, site coefficient for soil characteristics = 1.50, seismic zone coefficient = 0.25 (14)

Section – B

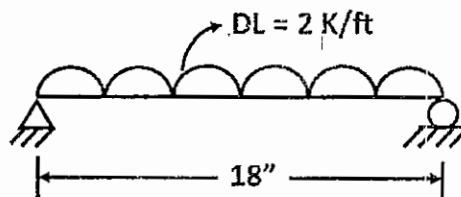
5. (a) Why deformed bar is used in reinforced concrete instead of plain bar? Differentiate between one way and two way slab. (08)
 (b) Differentiate between creep and shrinkage of concrete. (05)
 (c) If the T-beam shown in figure below is uncracked, calculate the stresses in the concrete at the top and bottom extreme fibers under a positive bending moment of 90 K.ft. What is the maximum uniformly distributed load the beam carry if it is used as a simple beam with 20 ft span without exceeding the modulus of rupture of the concrete. Given $f'_c = 3000$ psi. (22)



6. (a) Write down the assumptions made in Terzaghi's analysis for bearing capacity of soil under a strip footing. (05)
 (b) What is meant by foundation? Classify the general types of foundation with neat sketches. (13)
 (c) The SPT results at various depths in a soil are shown in table below. Determine the allowable bearing capacity for a square footing of 2.5 m wide located at a depth of 1.0 m below the ground level. Neglect ground water effect. Assume $\gamma = 20$ kN/m³. (17)

Depth (m)	0.5	0.8	1.1	1.4	1.7	2.0
N - value	9	12	11	13	14	17

7. (a) What is column? Show different types of columns with neat sketches. (10)
 (b) Design a square tied column for an axial load of 320 kips. Use $f'_c = 4$ ksi and $f_y = 50$ ksi. Assume $\rho_s = 0.02$. (25)
 8. (a) Design a rectangular section for the beam and loads shown in figure below. Live load is to be placed where it will cause the most severe condition at the section being considered. Given LL = 20 K, $f'_c = 4$ ksi and $f_y = 60$ ksi. (18)



- (b) A slab of 5 ft width is to be designed to place over a canal of width 15 ft. Given dead load is 100 psf including self weight and live load is 40 psf. Assume $f'_c = 5$ ksi and $f_y = 40$ ksi. Use WSD method. (17)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning

BURP 3rd Year 2nd Term Regular Examination, 2017

CE 3261

(Water Resources Planning)

Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Assume reasonable value for any missing data.

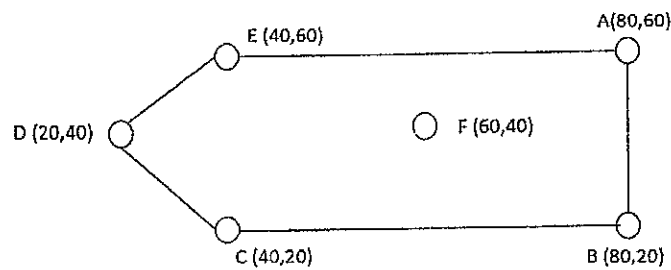
Section – A

1. (a) Define irrigation. What are the objectives of irrigation? (09)
(b) What are the qualities required for irrigation water? (10)
(c) What do you mean by safe perennial yield of ground water? (06)
(d) After how many days will you supply water to soil in order to ensure sufficient irrigation of the given crop, if (10)
(i) Field capacity of the soil = 26%
(ii) Permanent wilting point = 12%
(iii) Dry density of soil = 1.3 gm/cc
(iv) Effective depth of root zone = 60 cm
(v) Daily consumptive use of given crop = 10 mm
2. (a) Define regime channel. Design a regime channel for a discharge of 45 cumec and silt factor = 1.1, using Lacey's theory. (15)
(b) What is meant by 'Crop Period' and 'Base Period'? Show the relation between duty and delta. (10)
(c) Write short notes on – (10)
(i) Braided River, (ii) Effective Rainfall, (iii) Permanent Wilting Point, and
(iv) True Regime Channel
3. (a) Draw a neat sketch of an ideal river from source to outfall. Describe the division of fluvial system according to Schumm. (12)
(b) The following hydraulic data pertains to a bridge site of a river – (20)
Maximum discharge = 7000 cumec
Highest flood level = 105.00 m
River bed level = 100.00 m
Average diameter of river bed material = 0.12 mm
Design and sketch Bell's Bund including the launching apron to train the river.
Assume plentiful availability of boulders near the site.
(c) What is aggrading river? (03)
4. (a) What is meant by River Training Work? What are the objectives of River Training Works? (08)
(b) Write short note on : (20)
(i) Groyne, (ii) Guide Banks, (iii) Artificial Cut-Offs, and (iv) Pitched Island
(c) Define Return Period. Describe briefly the IDF Curve. (07)

Section – B

5. (a) What is meant by hydrology and hydrologic cycle? Describe hydrologic cycle with neat sketch. (12)
(b) Describe different forms of precipitation. (06)
(c) Write short note on : (05)
(i) Hyetograph
(ii) Mass Curve of Rainfall

- (d) Find the mean precipitation for the area sketched below by Thiessen Polygon method. Values in the paranthesis represents the ordinates in km. (12)



Station	A	B	C	D	E	F
Rainfall (cm)	7.0	6.0	4.8	7.1	6.8	7.5

6. (a) What is meant by Evaporation? Describe energy – budget method for determination of the lake evaporation. (10)
- (b) What is meant by Infiltration, Field Capacity and Infiltration Capacity? Describe with a simple analogy. (07)
- (c) The data pertaining to a stream – gauging operation at a gauging site are given below. The rating equation of the current meter is, $V = (0.51 N_s + 0.03)$ m/s. Calculate the discharge in the stream. (18)

Distance from Right edge (m)	0	1.0	4.0	7.0	9.0	11.0	14.0	17.0	20.0
Depth (m)	0	1.2	2.0	2.8	2.9	2.5	2.3	1.1	0
Revolutions of Current meter kept at 0.6 m depth	0	32	57	80	100	90	65	33	0
Duration of observation (min)	0	1.5	1.5	1.5	3.0	1.5	1.5	1.5	0

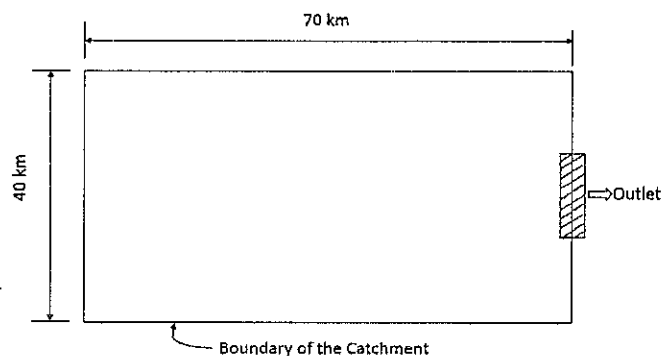
7. (a) Draw a typical hydrograph and show its different components. Define unit hydrograph. (10)
- (b) What is meant by excess rainfall? Describe different methods of base flow separation from a hydrograph. (12)
- (c) List the names of different factors that affect the shape of the hydrograph. Write down the assumptions involved in the unit hydrograph theory. (13)

8. (a) Differentiate between (08)
- (i) Base flow and Direct Runoff
 - (ii) Aquifer and Aquiclude
 - (iii) Specific Yield and Specific retention
 - (iv) Drawdown and Cone of depression

- (b) For the confined flow into a well show that, (12)

$$Q = \frac{2\pi K B S_w}{\ln \left(\frac{R}{r_w} \right)}$$

- (c) Derive a 4-hr unit Hydrograph for an ungauged basin as in figure below: (15)





Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2017
URP 3231
(Transportation Planning)

Full Marks: 210

Time: 3.0 hrs

- NOTE:** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) What is the purpose of trip generation? Explain the concept of aggregate and disaggregate model of trip generation in sequential travel demand forecasting model. (08)
- (b) In a sustainable urban development project what factors should be considered for ensuring walkable environment? In your opinion for walkable environment in Khulna city which two factors are most important and why so? (10)
- (c) Consider an urban road with an effective width of 8 feet including side obstruction of 2.5 feet. If the road operates at the breakpoint between LoS D and LoS E. Find out the linear capacity of this road. (07)
- (d) Find out the time-space demand and supply for pedestrian circulation of a Mezzanine of Metro station from the following information: (10)

Types of Movement (per hour)	Number of Persons	Time (Sec)
Stair to turnstiles	3200	15
Stairs to token booth to turnstiles	600	30
Stairs to newsstand to turnstiles	200	60
Turnstiles to newsstand to stairs	100	45

Assume that the area of the mezzanine is 1500 sq.ft including an unused area of 350 sq.ft and expected LoS is D.

2. (a) Make a comparative overview among Curb-side bus stop, Bus stop at a lay-by and Bus cape with necessary sketches. (12)
- (b) Exemplify different types of bus route with necessary sketches. Mention the principles of a bus route planning. (12)
- (c) What are the factors that influence the required bus fleet strength of a city? Calculate the required number of buses for Khulna city from the following data set: (11)
Population: 1,046,341; Per Capita Trip Rate: 1.5; Share of bus trips in overall trips: 15%; Average trip length: 8.2 Km; Distance travelled by bus per day: 200 km; Capacity of Bus: 65 passengers; Average occupancy: 0.8; Expected fleet utilization=90%.
3. (a) What is meant by time-mean speed and space-mean speed? Explain with necessary examples. (10)
- (b) What purpose moving observer method is used for? Illustrate the theory behind moving observer method. (15)
- (c) It is observed that on an average a vehicle driver drives 5000 km during the course of a year. The probability of having an accident is 100 per 200 million vehicle-kilometers. What is the probability of a driver having at least two accidents during his driving career extending to 25 years? (10)
4. (a) Using simple queuing theory derive: $P_n = \rho^n(1 - \rho)$, where ρ means traffic intensity, n means number of items and P_n means possibility that the queuing system has n items at a given time. (The service pattern is exponentially distributed and the flow has a poissonian arrival pattern). (15)
- (b) A toll booth at the entrance to a bridge can handle 120 veh/hour, the time to process a vehicle being exponentially distributed. The flow is 90 V.P.H with a poissonian arrival pattern. Determine: (i) the average number of vehicles in the system; (ii) the length of the queue; (iii) the average time spent by the vehicle in the system; (iv) the average time spent by the vehicle in the queue. (10)
- (c) Mention the factors should be considered while designing a road with a view to lessening road accidents. (10)

Section – B

5. (a) Explain the planning factors that are essential for a long term transportation planning? (08)
- (b) Make a list of different transportation planning studies. Write a short note on "Inventories" (10) and "Traffic Impact Studies" with examples.
- (c) Total number of trips from zone i to zone j is 20000. Currently all trips are made by car. (17) Government has two alternatives- to introduce a train or a bus. The travel characteristics and respective coefficients are given in the table.

	Vehicle travel time between 'i' and 'j'	Walking time to and from stops	Waiting time at stops	Fare charged to travel between 'i' and 'j'	Parking cost
Co-efficient	0.05	0.04	0.07	0.2	0.2
Car	21	-	-	25	7
Bus	30	10	5	8	-
Train	18	15	4	6	-

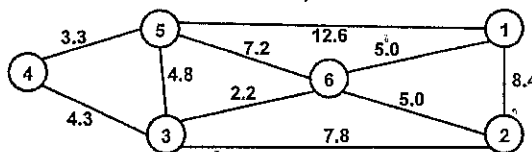
Decide the best alternative in terms of trips carried.

6. (a) Describe the factors that affect urban travel demand with examples. (08)
- (b) What are the different types of traffic a transportation planner deals with while planning for future travel demand management? (07)
- (c) Mention the selection criteria of Travel Analysis Zone (TAZ). (05)
- (d) The total trips produced in and attracted to the three zones A, B and C of a survey area in the design year are tabulated as: (15)

Zone	Trip Produced	Trip Attracted
A	2000	3000
B	3000	4000
C	4000	2000

If the trip interchange between zones B and C is known to be 600, calculate the trip interchange between zones A and B, A and C, B and A, C and B. Travel time between zones is uniform which is 20 minutes.

7. (a) Critically evaluate the importance of traffic assignment in transport planning? (10)
- (b) Write a short note on Smart Growth as an efficient TDM measure. (08)
- (c) The figure represents travel times on the link connecting six zones. Determine the minimum path between zones. Determine total trips for each link after all of the trips from the following two-way trip table have been loaded onto the network using all-or-nothing assignment method. (17)



Trips Between Zones						
From/To	1	2	3	4	5	6
1	0	1000	1100	400	1000	1300
2	-	0	1050	700	1100	1200
3	-	-	0	1200	1150	1600
4	-	-	-	0	800	400
5	-	-	-	-	0	700
6	-	-	-	-	-	0

8. (a) Describe the key requirements for Non-Motorized Transport (NMT) planning in Bangladesh. (12)
- (b) What are the differences between TSM and TDM? (05)
- (c) "Constructing a number of new flyovers is the right way to facilitate traffic flow and manage travel demand" - Do you agree with this statement? Justify your answer. (10)
- (d) Do you think Intermodal Coordination is an effective measures of travel demand management? Give a logical explanation with appropriate examples. (08)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2017
URP 3223
(Regional Development Planning)



Full Marks: 210

Time: 3 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

1. (a) Exemplify the subjective and objective view of regions. (10)
(b) Briefly discuss the importance of studying regional development planning for the planning students of Bangladesh. (10)
(c) How will you delineate a functional region using Flow Analysis and Gravitational Analysis? Explain with appropriate examples. (15)
2. (a) The resultant formal region will never be perfectly homogenous, but must be homogenous within certain clearly defined limits. Illustrate this statement with facts and figures. (15)
(b) Find the gravitational potential of a centre i from the following information, where i is surrounded by the centers A, B & C. (08)

Centre	Employment	Distance from i (m)
A	1000	100
B	1500	95
C	2000	210

Here, the value of $k = 1.5$

- (c) Define Economic Base Theory. Illustrate the limitations of this theory. (12)
3. (a) Mention the contribution of shift-share components to industrial structure analysis. (15)
(b) Exemplify the indirect approaches used for classifying the basic & non-basic activities. (15)
(c) The South-East Economy is dependent on two industries only: A, employing 5 million people at the beginning of the period, and B, employing 3 million, and both are growing faster than the national rate of approximately 6%, A at 8% and B at 11%. Now, calculate the proportionality shift component for this economy. (05)
4. Write short notes (any five): (35)
 - (a) Formal Region
 - (b) Functional Region
 - (c) Regionalization
 - (d) Regionalism
 - (e) Factors affecting regional disparities of Bangladesh
 - (f) Criteria of an administratively viable region by Smith
 - (g) Inter-Regional Trade Multiplier

Section – B

5. (a) Define Regional Planning. (08)
(b) "Regional Planning is an extension of Local Planning". Explain with appropriate example. (12)
(c) Shortly describe the importance of regional data base in regional planning. (15)
 6. (a) Clarify the concept of "Treble Entry" system. (05)
(b) Describe a simple input-output table with proper example for one area. (15)
(c) "Transport cost is directly proportional to distance moved and weight carried"- Explain with proper example. (15)
 7. (a) What is the basic concept of "Growth Pole Theory". (05)
(b) What is understood by "Central Place Theory"? Discuss about the three principles of "Central Place Theory" with necessary sketches. (20)
(c) Mention the shortcomings of aggregate growth models. (10)
 8. Write short notes on the followings (answer any five): (35)
 - (a) Leontief inverse matrix
 - (b) Inter Regional Resource Allocation Model
 - (c) Backwash in Economics
 - (d) Foot loose Industry
 - (e) Localization Economics
 - (f) Economic Space
 - (g) Limitations of input-output analysis
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Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 3rd Year 2nd Term Regular Examination, 2017
URP 3291
(Operations Research and Systems Analysis)

Full Marks: 210

Time: 3.0 hrs

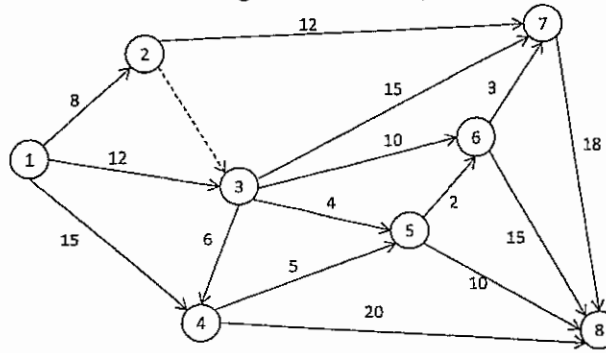
- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.
iii) Necessary graph papers will be supplied.

Section – A

1. (a) What is meant by Operations Research (OR)? Write down the characteristics of OR? (10)
- (b) An airline agrees to charter planes for a group. The group needs at least 160 first class seats and at least 300 tourist class seats. The airline must use at least two of its model 314 planes which have 20 first class and 30 tourist class seats. The airline will also use some of its model 535 planes which have 20 first class and 60 tourist class seats. Each flight of a model 314 plane costs the company TK. 1 lacs and each flight of a model 535 plane costs TK. 1.5 lacs. Formulate the problem as a linear programming model to find the number of each type plane should be used to minimize the flight cost? (15)
- (c) Mr. Nitrogen discovers that 700 square feet of plywood scrap and 480 square feet of white pine scrap are in usable form for the construction of tables and book cases. It takes 16 square feet of plywood and 8 square feet of white pine to make a table, 12 square feet of plywood and 16 square feet of white pine are required to construct a book case. By selling the finished products to a local furniture store Mr. Nitrogen can realize a profit of TK. 250 on each table and TK. 200 on each book case. Formulate the problem as a linear programming model to determine how may he most profitably use the left over wood. (10)
2. (a) Solve the following problem by simplex method. (20)
Minimize, $Z = 3x_1 + 5x_2$
Subject to, $x_2 \leq 6$
 $3x_1 + 2x_2 \leq 18$;
 $x_1, x_2 \geq 0$.
- (b) While solving a linear programming problem by simplex method what indicates by each of the following situation? (15)
(i) Infeasible solutions; (ii) Degeneracy; (iii) Unbounded solutions and (iv) Alternative optima
3. (a) What is meant by network analysis? Explain. (06)
- (b) Define the following terms: (09)
(i) Activity; (ii) Event; (iii) Dummy Activity
- (c) The activities involved in a project are listed in the following table. Construct the project network. (10)

Activity	Predecessor (s)	Duration (days)
A	-	3
B	-	5
C	-	2
D	A, B	7
E	B	10
F	B	5
G	B, C	4
H	B, C	12
I	D, E	6
J	F, G, I	3
K	D, E	8

- (d) Determine the critical path for the following network of a project. (10)



- 4 (a) What is simulation? Write down the methodology of a simulation study. (08)
 (b) What are the advantages and limitations of simulation? (07)
 (c) Dr. STRONG is a dentist who schedules all her patients for 20 minutes appointments. Some of the patients take more or less than 20 minutes depending on the type of dental work to be done. The following summary shows the various categories of work, their probabilities and the time actually needed to complete the work. (20)

Category	Time Required (min)	Probability of Category
Filling	25	0.10
Crown	35	0.35
Cleaning	15	0.25
Extraction	20	0.10
Check up	10	0.20

Simulate the dentist's clinic for two hours and determine the average waiting time for the patients. Assume that all the patients show up at the clinic at exactly their scheduled arrival time starting at 9.00 A.M. Use the following random number for handling the above problem.

84	57	14	43	72	4	68
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Section – B

5. (a) Define decision under risk. Explain it with suitable example. (08)
 (b) Consider the following pay-off (profit) matrix (10)

	Θ_1	Θ_2	Θ_3	Θ_4	Θ_5
a_1	7	9	-3	5	10
a_2	0	2	-3	5	10
a_3	14	27	39	-3	42
a_4	-13	30	19	47	2

No probabilities are known for the occurrence of natural states. Determine the solutions by the following criteria:

- (i) Laplace; (ii) Maximin; (iii) Savage; (iv) Hurwicz (assume $\alpha = 0.65$)
 (c) The oil India Corporation is considering whether to go for an off-shore oil drilling contract to be awarded in Bombay High. If they bid, value would be TK. 600 million with a 65% chance of gaining the contract. They may set up a new drilling operation or move already existing operation, which has proved successful, to the new site. The probability of success and expected returns are as follows: (17)

Outcome	Probability	New Drilling Operation Expected Revenue (TK. Million)	Probability	Existing Operation Expected Revenue (TK. Million)
Success	0.75	800	0.85	700
Failure	0.25	200	0.15	350

If the Corporation do not bid or lose the contract, they can use the TK.600 million to modernize their operation. This would result in a return of either 5% or 8% on the sum invested with probabilities 0.45 and 0.55 (Assume that all costs and revenue have been discounted to present value)

- (i) Construct a decision tree for the problem showing clearly the course of action.
 (ii) By applying an appropriate decision criterion recommend whether or not the oil India Corporation should bid the contract.

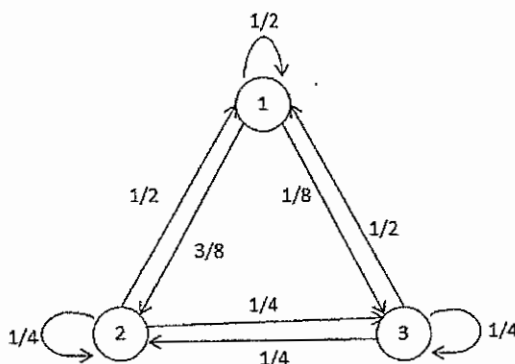
6. (a) Define mixed strategy. What is the role of mixed strategy in solving a game? (07)
 (b) Find the saddle point of the following game. (10)

		B			
		I	II	III	IV
A	I	4	-4	-5	6
	II	-3	-4	-9	-2
	III	6	7	-8	-9
	IV	7	3	-9	5

- (c) Solve the following game graphically. (18)

		B			
		I	II	III	IV
A	I	2	2	3	-1
	II	4	3	2	6

7. (a) Define queuing theory. Write down the managerial applications of queuing theory. (05)
 (b) Trucks of a company arrive at a transshipment centre to be unloaded in a pattern which is characterised by the poison distribution. The average rate of arrival is 45 per hour. A set of attendants unload the trucks, and the level of service is 100 per hour on an average. The drivers make BDT 16 per hour and the set of attendants are paid BDT 10 per hour. (14)
 (i) How much expense, on the average, is incurred by the company for idle time on part of each drivers each time he is at the transshipment centre?
 (ii) Find the optimum number or sets of attendants to be employed for transshipment.
 (c) A railway station has only one loading bay manned by a maximum of three - person crew. Studies identified that average arrival rate of trains for unloading is 4/hour and average service rate is 6 train/ hour. Explain the appropriateness of the number of crew in the system. The train waiting charges are BDT 20 per hour and the loading crew costs BDT 6 per hour. It can be assumed that work capacity is proportional to the number of crews. (16)
8. (i) What is Markov chain? Write down the application. (05)
 (b) A discrete process has the state diagram shown below. Find the probabilities of being in each of the three states after three steps using a tree diagram given that the process started in state 1. Determine the limiting state probabilities and the mean number of steps to enter state 3 if the process starts each time in state 2. (18)



- (c) A man's exercise habits are as follows. If he does his exercise one day, he is 70% sure not to do them the next day. On the other hand, if he does not do his exercise on day he is 60% sure not to do them the next day. In the long run, how often does he do his exercise? (12)